

DaSEH Day 2 Cheatsheet

Reproducibility

Major concepts

- **Reproducibility/Reproducible:** A different analyst re-runs the analysis with the same code and data and gets the same result.
- **Repeatable:** The same analysis is repeated under the same conditions and gives the same results.
- **Replicable:** New data are used to test whether the same inferences hold.

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	<code>sessionInfo()</code>	<code>sessionInfo()</code>	Returns the R version, OS, and attached packages in the current R session.
Base R	<code>sample()</code>	<code>sample(x, size = 10, replace = FALSE)</code>	Returns a random vector of a specified size, with or without replacement.
Base R	<code>set.seed()</code>	<code>set.seed(1234)</code>	Makes random-number output reproducible for subsequent code.

More resources

- [The R Markdown book](#)
- [Jenny Bryan's organizational strategies](#)
- [Write efficient R code for science](#)
- [Reproducibility in Cancer Informatics course](#)

Data Input

Major concepts

- **Delimited file:** A file in which columns are separated by punctuation, such as commas in CSV files or tabs in TSV/TXT files.
- **Tibble:** A rectangular data frame with data organized into rows and columns.
- **File path:** A file's location on your computer or the internet. [File paths can be relative or absolute.](#)
- **R Projects:** Set the working directory to the folder containing the `.Rproj` file.
- **Reading in data manually:** Download the data and put it in the project folder. In RStudio, go to File > Import Dataset > From Text (`readr`), browse for the file, then click "Update" and "Import."

Functions

Library/Package	Piece of code	Example of usage	What it does
tidyverse (<code>readr</code>)	<code>read_csv()</code>	<code>df <- read_csv("<url>")</code>	Reads in a CSV file from a specified file path.
tidyverse (<code>readr</code>)	<code>read_tsv()</code>	<code>df <- read_tsv("<url>")</code>	Reads in a TSV file from a specified file path.
tidyverse (<code>readr</code>)	<code>read_delim()</code>	<code>df <- read_delim("<url>", delim = ":")</code>	Reads in a delimited file from a specified file path.
Base R	<code>head()</code> / <code>tail()</code>	<code>head(df)</code> / <code>tail(df)</code>	Returns the first or last part of a dataset or other object.
Base R	<code>getwd()</code>	<code>getwd()</code>	Returns the current working directory.
Base R	<code>setwd()</code>	<code>setwd("Desktop")</code>	Changes the current working directory.
Base R	<code>View()</code>	<code>View(df)</code>	Opens a spreadsheet-style data viewer for a matrix-like R object.
Base R	<code>str()</code>	<code>str(df)</code>	Displays the internal structure of an R object, including dimensions and class.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.